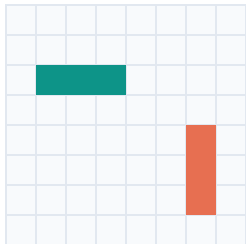
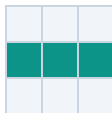


12-16 · 2-D: two operations need two channels — and the law breaks at three

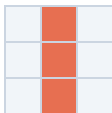
task: both bars present?



two channels = two detectors

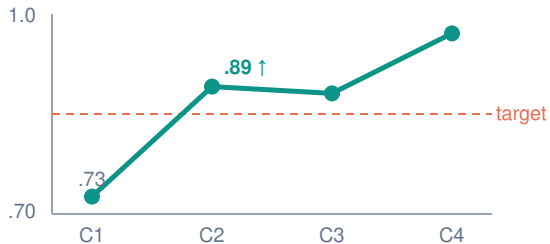


horizontal



vertical

accuracy vs channels (2-op task)



one op → 1 channel · two ops → 2 channels (C=1 caps at the 0.75 a single detector can reach)

but the tidy “width = operations” law BREAKS at K=3: needs redundant channels, and three rescues (more channels, MLP head, 2nd conv layer) all fail